SGA 2: Application to the Fundamental Group

Purity theorem — proof of part (i)

Bounded source-aligned working unit — 22 July 2026

Authority and boundary. Corrected French TeX lines 3568–3571. The unit begins on original printed p. 121 and continues on printed p. 122; both portions occur on source-PDF physical p. 105 (recomposed running p. 97). Blank line 3572 is excluded. The raw continuation cursor is line 3572 and the next substantive cursor is line 3573, which begins part (ii) and is excluded in full. The French authority remains byte-identical. The same-edition PDF is manifestation and layout evidence only, and the external English candidate is comparison only. Independent review and archive handoff have not been performed; this package is internal and not for release.

Inherited-context observation [SGA2-X-L3569-REGULARITY-CONTEXT-OBS-001]. Theorem 3.4(i), whose proof begins here, concerns regular noetherian local rings. French line 3569 initially says only “a noetherian local ring of dimension 2” and later explicitly invokes its regularity. This translation therefore retains the printed wording while making clear that regularity is inherited from the theorem being proved; it does not introduce a new line-3569 hypothesis.

Proof of the purity theorem. We first prove (i) by induction on the dimension. Let A be a noetherian local ring of dimension 2. Set $X' = \operatorname{Spec}(A)$, $x = \mathfrak{r}(A)$, and $X = X' \setminus \{x\}$. We have $\operatorname{depth} A = 2$. We may therefore apply Lemma 3.5 to the pair $(X', \{x\})$, and hence $\mathbf{Et}(X') \longrightarrow \mathbf{Et}(X)$ is fully faithful.

Now let $r\colon R \longrightarrow X$ be an étale covering defined by the coherent, locally free, étale $\mathcal{O}_X$-algebra $\mathcal{A} = r_*(\mathcal{O}_R)$. Let $i\colon X \longrightarrow X'$ denote the canonical immersion. I claim that $i_*(\mathcal{A}) = \mathcal{B}$ is a coherent $\mathcal{O}_{X'}$-algebra. Indeed, it suffices to apply the “finiteness theorem,” VIII, 2.3. I also claim that this algebra has depth at least 2 at x: it is the direct image of an $\mathcal{O}_X$-module, with $X = X' \setminus \{x\}$. Since A is a regular ring of dimension 2, we have

$$\operatorname{pd} \mathcal{B} + \operatorname{depth} \mathcal{B} = \dim A = 2,$$

where $\operatorname{pd} \mathcal{B}$ denotes the projective dimension of $\mathcal{B}$. Thus $\operatorname{pd} \mathcal{B} = 0$, so $\mathcal{B}$ is projective and therefore free. Consequently, $\mathcal{B}$ defines a finite flat covering of $X' = \operatorname{Spec}(A)$. The set of points of X' at which this covering is not étale is a closed subset of X' whose defining ideal is principal, namely the discriminant ideal of $\mathcal{B}/A$. By construction this closed subset is contained in $x = \mathfrak{r}(A)$; it is therefore empty because $\dim A = 2$.

Let A now be a regular noetherian local ring with $\dim A = n \geq 3$. Suppose that (i) has been proved for rings of dimension less than n. To prove that A is pure, we may assume by Corollary 3.8 that A is complete. Choose $t \in \mathfrak{r}(A)$ whose image in $\mathfrak{r}(A)/\mathfrak{r}(A)^2$ is nonzero. Then $B = A/tA$ is a regular noetherian local ring of dimension $n - 1$, and is therefore pure because $n - 1 \geq 2$. The conclusion follows from Lemma 3.9(i), which is applicable because A is complete.